Characterization of Quasi Stationary Distribution in a long-range interacting system

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Abstract

Hamiltonian mean-field model (HMF model) is a mean-field N-pendulum system, which is introduced to understand the behavior of systems with a long-range interaction. The system exhibits various quasi stationary states (QSS) according to initial condition. We introduce moment-like variable set, in order to characterize QSS. We show that each variable is decided by limited set of variables. It indicate the possibility that QSS is controlled by limited number of variables.

Keyword: Hamiltonian dynamics, Long-range interactions

Hamiltonian mean-filed (HMF) model is a mean-field N-pendulum system, which is introduced to understand the behavior of systems with a long-range interaction. HMF model is and has non-thermodynamical stationary states (QSS). The Hamiltonian is expressed as $H=\sum_{i=0}^{n}\frac{p^2}{2}+\frac{1}{2N}\sum_{i.j=1}^{n}[1-\cos(\theta_i-\theta_j)]$. In the thermodynamical limit, the system cannot escape from QSS[1]. Lynden-Bell theory seems to describe the magnetization curve well[2], but the order of transition is not correct. Utilizing Core-Halo one-body distribution describes the magnetization M curve better and the transition order agrees with the N-body simulation[3].

We investigate how the one-body phase distribution of QSS changes as the initial condition. In some cases, distribution which is not reported is realized. To characterize QSS distribution, we measured moment-like variables μ_i. We find functional relationship between them in some case. It indicate the possibility that QSS is controlled by limited number of variables.

References

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